

## Chapter 2 - Summary and Sample Exam Questions

### Why a VAE is needed

A deterministic autoencoder learns a mapping from an input  $x$  to a latent code  $z$ , and then back to a reconstruction  $\hat{x}$ . This is useful for compression and denoising, but it is **not** a principled generative model. The latent codes produced by a standard autoencoder can occupy arbitrary disconnected regions of latent space, so drawing a random latent vector and decoding it often produces unrealistic outputs. In other words, a deterministic autoencoder can reconstruct training-like examples, but it does not tell us what probability model generated the data.

A Variational Autoencoder fixes this by introducing a **probabilistic latent-variable model**. Instead of saying that each input maps to one exact latent vector, the encoder produces a **distribution** over latent variables. The decoder is also probabilistic: it defines a likelihood for the data conditioned on the latent variable. This gives the model a proper generative story.

### The probabilistic model behind a VAE

A standard VAE assumes:

$$p(z) = \mathcal{N}(0, I)$$

and

$$p_\theta(x \mid z)$$

where  $p(z)$  is the prior over latent variables and  $p_\theta(x \mid z)$  is the decoder or generative model. The joint model is:

$$p_\theta(x, z) = p_\theta(x \mid z) p(z)$$

and the marginal likelihood of the observed data is:

$$p_\theta(x) = \int p_\theta(x \mid z) p(z) dz$$

The goal is to learn  $\theta$  so that  $p_\theta(x)$  matches the true data distribution. The challenge is that the integral over  $z$  is generally intractable when the decoder is a neural network. This is the central computational problem of VAEs.

### Why variational inference appears

If we could compute the true posterior

$$p_\theta(z \mid x) = \frac{p_\theta(x \mid z) p(z)}{p_\theta(x)}$$

then learning and inference would be much easier. But this posterior depends on  $p_\theta(x)$ , which itself requires the intractable integral above. VAEs therefore introduce an approximate posterior, also called the encoder:

$$q_\phi(z \mid x)$$

In practice, this is usually chosen as a Gaussian with diagonal covariance:

$$q_\phi(z \mid x) = \mathcal{N}(\mu_\phi(x), \text{diag}(\sigma_\phi^2(x)))$$

where the encoder network outputs  $\mu_\phi(x)$  and  $\log \sigma_\phi^2(x)$ .

The key idea is not that  $q_\phi(z \mid x)$  is the true posterior, but that it is a **tractable family** of distributions that can be optimized jointly with the decoder.

### The ELBO and what it means

Because  $\log p_\theta(x)$  is hard to optimize directly, the VAE maximizes a lower bound called the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\text{ELBO}}(x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - \text{KL}(q_\phi(z \mid x) \parallel p(z))$$

This objective has two conceptually different parts.

The first term is the **reconstruction or likelihood term**. It encourages the latent variable inferred from  $x$  to contain enough information so that the decoder can explain or reconstruct  $x$  well.

The second term is the **regularization term**. It pushes the approximate posterior toward the prior. This makes the latent space smoother, more organized, and more sampleable. Without this pressure, the model could behave like a deterministic autoencoder with arbitrary latent codes.

The ELBO is not just a heuristic. It satisfies:

$$\log p_\theta(x) = \mathcal{L}_{\text{ELBO}}(x) + \text{KL}(q_\phi(z \mid x) \parallel p_\theta(z \mid x))$$

Since the KL divergence is always nonnegative, the ELBO is indeed a lower bound on the log marginal likelihood. Maximizing the ELBO therefore does two things at once: it increases a tractable surrogate for model evidence, and it makes the approximate posterior closer to the true posterior.

### Why reparameterization matters

The ELBO contains an expectation over samples from  $q_\phi(z \mid x)$ . If one samples  $z$  directly from a distribution whose parameters depend on  $\phi$ , naive sampling breaks standard backpropagation. The reparameterization trick solves this by isolating the randomness in a parameter-free noise variable:

$$\epsilon \sim \mathcal{N}(0, I)$$

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon$$

Now the stochasticity comes from  $\epsilon$ , while  $z$  is a differentiable function of the encoder outputs. This lets gradients flow through the sample and enables efficient end-to-end learning with gradient descent.

Conceptually, reparameterization changes the question from "how do we differentiate through random sampling?" to "how do we express sampling as a deterministic transformation of fixed noise?" This is one of the core innovations that makes VAEs practical.

### Geometry, information, and failure modes

A well-trained VAE learns a latent space in which nearby points tend to decode to similar outputs. This is what makes interpolation and random sampling meaningful. However, the latent space is shaped by a tension: too little regularization gives poor generative structure, while too much regularization can destroy information needed for accurate reconstruction.

This trade-off explains several important variants and failure modes.

In a  **$\beta$ -VAE**, the KL term is weighted more strongly:

$$\mathcal{L}_{\beta\text{-VAE}}(x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - \beta \text{KL}(q_\phi(z \mid x) \parallel p(z))$$

When  $\beta > 1$ , the model is encouraged to use a more factorized and compressed latent code, which can promote disentanglement. But if  $\beta$  is too large, reconstructions degrade because the latent bottleneck becomes too restrictive.

A second extension is the **Conditional VAE (CVAE)**, which conditions both encoder and decoder on known information  $y$ :

$$q_\phi(z \mid x, y), \quad p_\theta(x \mid z, y)$$

This is useful when part of the variation is known in advance, such as a class label. Conditioning reduces the burden on  $z$ : instead of encoding everything, the latent variables only need to model residual variation. This often improves controllability and reduces entanglement.

A common failure mode is **posterior collapse**, where:

$$q_\phi(z \mid x) \approx p(z)$$

for many or all inputs. In that case,  $z$  carries little information about  $x$ , and the decoder effectively ignores the latent variable. This often happens when the decoder is too powerful or when KL pressure is too strong early in training.

### Summary

A VAE is best understood as a bridge between **deep learning** and **probabilistic inference**. It combines:

- a latent-variable generative model,
- approximate Bayesian inference through  $q_\phi(z \mid x)$ ,
- and end-to-end neural optimization via reparameterization.

Its deeper conceptual significance is that representation learning is no longer only about compressing data into features. It becomes about learning a latent space that supports **inference**, **generation**, **sampling**, and **controlled manipulation**. The VAE therefore occupies an important place in modern machine learning: it is both a generative model and a framework for learning structured probabilistic representations.

### Question 1

Why is the ELBO called a lower bound on  $\log p_\theta(x)$ , and under what condition does it become exact?

#### Sample Answer

The ELBO is called a lower bound because

$$\log p_\theta(x) = \mathcal{L}_{\text{ELBO}}(x) + \text{KL}(q_\phi(z \mid x) \parallel p_\theta(z \mid x))$$

Since KL divergence is always nonnegative, it follows that

$$\mathcal{L}_{\text{ELBO}}(x) \leq \log p_\theta(x)$$

for every datapoint.

The bound becomes exact when the approximate posterior matches the true posterior:

$$q_\phi(z \mid x) = p_\theta(z \mid x)$$

In that case the KL gap is zero. This shows that VAE learning is not only about learning a decoder; it is also about learning an inference model that makes the bound tight.

### Question 2

Interpret the two main terms in the ELBO,

$$\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] \quad \text{and} \quad \text{KL}(q_\phi(z \mid x) \parallel p(z))$$

and explain the tension between them.

#### Sample Answer

The expectation term rewards the model for explaining the observation well using latent values that are plausible for that same observation under  $q_\phi(z \mid x)$ . In practice, this plays the role of a probabilistic reconstruction objective.

The KL term regularizes the approximate posterior so that it does not drift too far from the prior. This encourages a more organized latent space that supports sampling and reduces the risk that each datapoint receives an isolated, arbitrary code.

The tension is that better reconstruction often pushes the encoder to store more information about  $x$  in the latent variable, while stronger KL pressure pushes the latent representation toward a simpler distribution. A VAE must balance fidelity to the data against structure and regularity in latent space.

### Question 3

Explain the reparameterization trick conceptually. Why is it more than just a coding convenience?

#### Sample Answer

The reparameterization trick rewrites a latent sample as

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

This separates randomness from the learnable parameters. The noise is drawn from a fixed source, while the encoder controls how that noise is transformed into a latent sample.

Conceptually, this matters because it makes the stochastic step behave like a differentiable computation. Gradients can then flow through the latent sample back into the encoder parameters. Without this idea, sampling would interrupt ordinary backpropagation and make learning much harder.

So the trick is not merely implementation detail. It is one of the key ideas that makes variational autoencoders trainable with standard deep learning methods.

### Question 4

A standard VAE does not automatically guarantee disentangled latent variables. Why not, and how do ideas such as  $\beta$ -VAE or CVAE try to change this?

#### Sample Answer

A standard VAE encourages reconstruction quality and approximate posterior regularity, but those pressures alone do not force separate latent coordinates to align with separate semantic factors. The model may still encode several factors together in entangled ways if that helps optimize the objective.

A  $\beta$ -VAE modifies the tradeoff by increasing the weight on the KL term:

$$\mathcal{L}(x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - \beta \text{KL}(q_\phi(z \mid x) \parallel p(z))$$

When  $\beta > 1$ , the model is pushed toward a more constrained latent representation, which can encourage factor separation, although often at the cost of reconstruction quality.

A CVAE introduces side information such as labels or attributes. By explicitly conditioning on known factors, it reduces the burden on the latent variables and can produce a more structured representation. In both cases, the broader lesson is that disentanglement usually requires extra inductive bias rather than emerging automatically.

### Question 5

Suppose a VAE encoder learns an approximate posterior that is almost identical to the prior for every input, so that

$$q_\phi(z \mid x) \approx p(z)$$

for all  $x$ . What does this suggest about how the model is using its latent variables, and what consequence would you expect for reconstruction quality?

#### Sample Answer

If  $q_\phi(z \mid x)$  is nearly the same as  $p(z)$  for every input, then the latent representation carries very little information about the specific datapoint  $x$ . In effect, the encoder is no longer distinguishing one input from another in a meaningful probabilistic way.

This usually means the decoder is being forced to reconstruct without relying much on the latent code. If the decoder is powerful enough, it may still produce reasonable outputs by learning generic patterns in the data, but reconstruction quality for individual examples will typically suffer because the latent variable is not providing input-specific information.

Conceptually, this reflects a failure to use the latent space as an informative hidden representation. The model may still optimize part of the objective, especially the KL term, but it is no longer achieving the intended balance between informative encoding and regularized generation.